YANG XU

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EDUCATION

University of Toronto, Ontario, Canada
MSc Computer Science — 2026 – Current

University of Toronto, Ontario, Canada
BASc Engineering Science, Robotics Major — 2021 – 2026

- **CGPA: 3.99/4.00**.
- **Courses:** deep learning, computer vision, data structures & algorithms, computer architecture, dynamics, control systems, electronics, robot modelling, variety of mathematics & physics.

RESEARCH EXPERIENCE

Toronto Intelligent Systems Lab - Research Intern — Ontario, Canada | May 2025 – May 2026
Advisor: Prof. Igor Gilitschenski
Project: Language-conditioned multi-embodiment robot grasp generation in cluttered scenes.

- Submitted to CoRL 2026.
- Engineered a multi-stage IsaacSim data-generation pipeline integrating cluttered scene synthesis, custom heuristics for grasp-stability and collision scoring, LLM-prompted image annotation, and distributed parallelization, yielding a dataset with millions of grasps across several robotic grippers.
- Developed and trained a flow-matching generative model that outputs grasp poses conditioned on the full scene point cloud representation.
- Developed a language-conditioning framework for grasp pose generation using zero-shot VLMs, removing the need for additional training and language data annotations.

WORK EXPERIENCE

Tenstorrent - Systems Engineer — Ontario, Canada | May 2024 – May 2025

- Designed and executed comprehensive test plans for AI processors at chip, board, and product levels, verifying performance and functionality against pre-silicon expectations and presenting detailed analysis to stakeholders.
- Defined V/F and power/perf. curves for next-gen AI processor by implementing an automated testing procedure, debugging ASIC firmware and PCB design, and optimizing performance within power and thermal constraints.
- Conducted SERDES and GDDR testing to assess signal integrity, analyzing bit error rates (BER) and eye diagram measurements to ensure optimal component performance.
- Implemented firmware features such as Built-In Self-Test (BIST) for SERDES and automated SERDES firmware loading during ASIC boot-up, streamlining initialization and verification processes.

Rocscience - Research Engineer — Ontario, Canada | May 2023 – Aug. 2023

- Conducted in-depth research of rigid-body impact mechanics with friction in 2D and 3D settings, aimed at enhancing the accuracy of rigid-body rockfall simulations used for mining and construction.
- Designed a 3D rock impact model grounded in Stronge's impact mechanics, capturing energy dissipation via friction and plastic deformation, and implemented numerical solutions using quadratic programming and fixed-point iteration.
- Implemented new collision resolution algorithm to the RocFall3 physics engine using C++, resulting in an estimated accuracy increase of 50% and an estimated practical performance improvement of 15%.

AMD - Computer Engineer — Ontario, Canada | May 2022 – Aug. 2022

- Researched and developed a 90% accurate machine learning model that scrubs gigabytes of raw data for useful power & performance analytics, generalizing across multiple computer configurations.
- Developed automation scripts and data-analysis tools like data parsers and data visualizers to streamline hardware experimentation and data analysis work.
- Conducted computer graphics experiments to explore new methods of improving CPU and GPU power allocation in Smart Shift to increase performance in future AMD-powered laptops.

SCHOLARSHIPS & AWARDS

- **2026 Queen Elizabeth II Graduate Scholarships in Science & Technology (QEII-GSST)** $15,000 CAD awarded by the Province of Ontario and the Unversity of Toronto for graduate studies.
- **2025 Undergraduate Student Research Award (USRA)** $7,500 CAD awarded by Natural Sciences and Engineering Research Council of Canada for undergraduate summer research project.
- **2024 University of Toronto Scholar** $1,500 CAD awarded to top academically performing students.
- **2021 University of Toronto Entrance Scholarship** $5,000 CAD.

LANGUAGES

English Native

Chinese Fluent

French Intermediate

TECHNICAL SKILLS

Proficient Python, PyTorch, C/C++, MATLAB, ROS, Verilog.

Familiar Linux, TensorFlow, Fusion 360, LaTeX.